

FPL-Oracle Redis Schema (Two-DB Architecture)

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This document describes the finalized, production-ready Redis schema for the FPL-Oracle system. It includes:

- Key namespaces
 - Redis data types
 - Example values
 - DB separation rules
-

1. Database Separation

FPL-Oracle uses **two Redis databases**:

- **DB 0 - RAW DATA**
 - All scraped, unprocessed, or minimally normalized data
 - All keys **without** the `proc_` prefix
- **DB 1 - PROCESSED DATA**
 - All cooked, computed, or model-derived data
 - All keys **starting with** `proc_`

Routing rule:

```
if key.startswith("proc_") → DB 1  
else → DB 0
```

2. DB 0 - RAW DATA

2.1 Players (RAW)

``player:`

Core identity + current FPL stats.

Example:

```
player:302
  name: "Bruno Fernandes"
  team_id: 13
  position: 3
  cost: 8.5
  status: "a"
  chance_of_playing: 100
```

`player:

Metadata from FPL.

```
player:302:meta
  selected_by_percent: 22.4
  transfers_in: 54000
  transfers_out: 12000
  news: "Available"
```

`player:

Past season totals (year normalized to end year).

```
player:302:season:2025
  minutes: 3100
  goals: 10
  assists: 8
  xG: 9.2
  xA: 7.5
```

`player:

Per-gameweek performance.

```
player:302:gw:38
  minutes: 90
  goals: 1
  assists: 0
  bps: 27
  points: 8
```

`player:

Per-fixture detailed stats.

```
player:302:fixture:373
  shots: 4
  key_passes: 3
  xG: 0.41
  xA: 0.28
  touches: 72
```

2.2 Teams (RAW)

`team:

```
team:13
  name: "Manchester United"
  short_name: "MUN"
  strength_overall_home: 78
  strength_overall_away: 74
```

`team:

```
team:13:expected
  xG: 1.62
  xGA: 1.21
  xPts: 1.44
```

`team:

```
team:13:form
  form_string: "WWDLW"
  goals_for: 9
  goals_against: 6
  points: 10
```

2.3 Fixtures (RAW)

`fixture:

```
fixture:373
  home_id: 8
  away_id: 1
  kickoff_time: "2026-05-18T14:00:00Z"
  finished: true
  score: "1-1"
```

`fixture:

```
fixture:373:stats
  shots_home: 12
  shots_away: 9
  xG_home: 1.41
  xG_away: 0.98
  possession_home: 54
```

2.4 Gameweeks (RAW)

`gw:

```
gw:38
  name: "Gameweek 38"
  deadline_time: "2026-05-18T11:00:00Z"
  is_current: false
  highest_score: 147
  most_captained: 430
```

`gw:

```
gw:38:fixtures = {373, 374, 375}
```

`gw:

```
gw:38:players = {302, 430, 147, ...}
```

`gw:

```
gw:38:teams = {1, 8, 13, ...}
```

2.5 Indexes (RAW)

`index:player_fixtures:

```
ZADD index:player_fixtures:302 38 373
ZADD index:player_fixtures:302 37 361
```

`index:team_fixtures:

```
ZADD index:team_fixtures:13 38 373
```

`index:team_name:

```
index:team_name:Manchester United
tid: 13
```

`index:team_players:

```
index:team_players:13 = {302, 415, 188}
```

`index:position_players:

```
index:position_players:3 = {302, 415}
```

`index:season_players:

```
index:season_players:2025 = {302, 430, 147}
```

`index:season_fixtures:

```
index:season_fixtures:2025 = {373, 374, 375}
```

2.6 System State (RAW)

system:state - HASH

```
system:state
  current_gw: 38
  current_season: 2025
  last_producer_run: "2026-05-18T12:00:00Z"
  cook_status: "complete"
```

2.7 Raw Dumps (RAW)

`raw:fpl:

```
"{ \"elements\": [...], \"teams\": [...], ... }"
```

`raw:fotmob:

```
"{ \"xG\": { ... }, \"stats\": [...], ... }"
```

3. DB 1 - PROCESSED DATA

All keys begin with proc_.

3.1 Players (PROC)

`proc_player:

```
proc_player:302:xp
  xp_this_gw: 7.4
  xp_next_gw: 5.1
  minute_probability: 0.91
  form_coefficient: 1.12
  fixture_difficulty: 2
  cs_probability: 0.34
  captain_score: 8.2
  computed_at: "2026-05-18T12:30:00Z"
```

3.2 Teams (PROC)

`proc_team:

```
proc_team:13:strength  
  attack_overall_expected: 1.12  
  defence_overall_expected: 0.94  
  point_overall_expected: 1.18
```

3.3 Fixtures (PROC)

`proc_fixture:

```
proc_fixture:373:poisson  
  0: 49.659  
  1: 28.305  
  2: 8.067  
  3: 1.533
```

`proc_fixture:

```
proc_fixture:373:scoreline  
  "0-0": 49.659  
  "1-0": 6.456  
  "0-1": 28.305  
  "1-1": 3.680
```

3.4 Gameweeks (PROC)

`proc_gw:

```
proc_gw:38:team_of_week  
  gk: 302  
  def_1: 155  
  def_2: 264  
  def_3: 188  
  mid_1: 427  
  fwd_1: 90  
  captain: 302  
  vice_captain: 427
```

3.5 Indexes (PROC)

proc:index:players - SET

```
proc:index:players = {302, 415, 188}
```

4. Summary

This schema provides:

- Clean two-DB separation
 - Predictable key namespaces
 - Correct Redis types
 - Fast O(1) lookups
 - Full support for raw + cooked pipelines
- It is now production-ready and suitable for FastAPI, Next.js, and future ML cooks.